27 Infection Control and Isolation Procedures

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27.1 Introduction

Infection prevention and control practices (IPC) are defined as a set of measures aimed at preventing or stopping the spread of infections in healthcare settings. All HCT recipients should follow general guidelines (e.g. provided by CDC) for preventing healthcare-associated infections through hand hygiene, disinfection and sterilisation, environmental infection control, isolation precautions and prevention of intravascular catheter-related infection [(Sehulster et al. 2004; Centers for Disease Control and Prevention 2002; Guideline for Isolation Precautions: Preventing Transmission of Infectious Agents in Healthcare Settings 2007; Guidelines for Environmental Infection Control in Health-Care Facilities 2003), all available at https://www.cdc.gov/infection-control/guidelines].

Dedicated and detailed international recommendations for HCT recipients on preventing infectious complications have been published in 2009 (Tomblyn et al. 2009; Yokoe et al. 2009). As there were no well-executed randomised or controlled trials and little evidence from cohort or multiple time-series studies, most data come from descriptive studies, reports of expert committees and the opinions of respected authorities. Hence, recommendations on infection control could only be graded as level III.

Isolation procedures in HCT recipient comprise precautions universal for all healthcare settings (Standard Precautions and Transmission-Based Precautions) and those specific for HCT and employed to prevent transmission of spores of filamentous fungi, mainly *Aspergillus*, with unfiltered air.

There is no consensus on specific *protective environment*, also called reverse isolation, for neutropenic patients. HCT recipients should be placed in a single-patient room, with adequate ventilation system (see below), if possible. However, no clear benefit of routine footwear exchange or use of disposable gloves and gowns on the rate of infections has been demonstrated, and procedures vary significantly between institutions, with routine use of masks and disposable gloves and gowns in some but not others. On the contrary, the negative effect of strict protective isolation on patient's quality of life and well-being should be acknowledged and weighted against the evidence of benefits of single protective measures (Abad et al. 2010).

27.2 Standard Precautions

Should be used universally for all patients and they include

1. Proper hand hygiene

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© The Author(s) 2024
A. Sureda et al. (eds.), *The EBMT Handbook*, https://doi.org/10.1007/978-3-031-44080-9_27

2. Use of standard personal protective equipment (PPE)
3. Appropriate cleaning and disinfection protocols (including those for shared equipment or toys and play areas in paediatric units)
4. Safe injection practices
5. Infection control practices for special procedures (e.g. surgical masks for lumbar puncture)

27.2.1 Hand Hygiene

It is by far the most effective means of prevention of pathogen transmission (Freifeld et al. 2011; Tomblyn et al. 2009). The preferred method of hand decontamination is with an alcohol-based hand rub, due to its superior convenience and reduced drying of the skin. Handwashing with soap and water is recommended if hands are visibly soiled, for example, with blood or body fluids, and after potential contact with spores of *Clostridioides difficile* or with *Norovirus*. Of note, 15–30 s is the minimum necessary handwashing time.

27.2.2 Other Standard Precautions

PPE used routinely by healthcare workers during patient care and procedures are gloves, gowns (used if direct contact with patient's fluids is expected) and mouth, nose and eye protection (used during procedures which are likely to generate splashes or sprays of blood, body fluids, secretions and excretions). Routine donning of gowns upon entrance into a high-risk unit, including HCT unit, is not indicated.

27.3 Transmission-Based Precautions

These are the measures used in addition to standard precautions for patients with documented or suspected infection or colonisation with highly transmissible or epidemiologically important pathogens for which additional specific precautions are necessary to prevent transmission. The main types of transmission-based precautions are *contact* precautions, *airborne* precautions and *droplet* precautions. The specific PPE and the examples of pathogens which require each type of transmission-based precautions are outlined in Table 27.1.

Transmission-based precautions should also be applied in a *pre-emptive* way, e.g. in case of patients transferred from high-risk facilities, pending the results of surveillance cultures or during diagnostic workup, e.g. for suspected infectious diarrhoea.

There are no clear criteria for appropriate discontinuation of contact precautions (usually multiple negative swabs from a site known to be colonised), but there are data suggesting that colonisation with multidrug-resistant (MDR) pathogens might persist longer and reappear after several negative swabs, and *C. difficile* shedding might be present even after the resolution of diarrhoea (Banach et al. 2018). Therefore, in HCT setting, continuing contact precautions until discharged home and reculturing patients to document clearance of MDR carriage only after an interval free of hospitalisations, antimicrobial therapy and invasive devices might be more appropriate (Sehulster et al. 2004; Banach et al. 2018). The information on contact precautions in place should be clearly stated on the discharge information form for the centres which will care for such patient subsequently. In case of MDR Gram-negative pathogens, full antibiotic susceptibility results should be provided to allow appropriate empirical therapy in case of severe subsequent infection. In case of *Candida auris* colonisation, no routine reassessment of colonisation is recommended since it may persist for many months and discontinuation of contact precautions in healthcare facility is not recommended (updated recommendations available at https://www.cdc.gov/fungal/candida-auris/c-auris-infection-control.html).

Cough etiquette should be promoted. Additionally, transplant recipients, particularly those with respiratory symptoms, should use surgical masks and maintain special separation from others in common waiting areas, ideally a distance of at least 1 m.

Table 27.1 Transmission-based precautions to be applied in addition to standard precautions

Type of precaution	Patients' placement and PPE to be used by patients	PPE for healthcare personnel	Example of pathogens and comments
Contact	– Single room; if not available, cohorting of those colonised/infected by the same pathogen – During transport, cover patient's colonised/infected areas	– Disposable gloves and gowns – Use patient-dedicated or disposable equipment; if not feasible, clean and disinfect thoroughly	– Infection with *Clostridioides difficile* – Colonisation or infection with MDR pathogens – Infectious diarrhoea due to pathogens such as *Salmonella*, *Norovirus*, *Rotavirus*, etc. All the units and other hospitals involved in patient's care should be notified about all the isolated pathogens requiring contact precautions
Droplet	– Single room; if not available, cohorting of those infected by the same pathogen – Surgical mask – Follow CDC's respiratory hygiene/cough etiquette in healthcare setting	– Mask (surgical) – In many cases in HCT also disposable gloves and gowns, since both droplet and contact precautions are necessary e.g. in case of respiratory viruses	– Pathogens transmitted by respiratory droplets (i.e. large-particle droplets >5 μ in size) that are generated by a patient who is coughing, sneezing or talking, e.g. influenza or other respiratory viruses In case of transplant recipients, the duration of precautions should be extended due to the possibility of prolonged shedding caused by immunodeficiency
Airborne	– Preferably airborne infection isolation room, i.e. with at least 6 (existing facility) or 12 (new construction/renovation) air changes per hour and direct exhaust of air to the outside (if not possible, the air may be returned to the air-handling system or adjacent spaces if all air is directed through HEPA filters) – Surgical mask – Follow CDC's respiratory hygiene/cough etiquette in healthcare setting	– N95 or higher-level respirator for respiratory protection	– *Mycobacterium tuberculosis* (patients with respiratory tuberculosis and sputum with direct evidence of mycobacteria) – Measles, chickenpox and disseminated herpes zoster

HEPA high-efficiency particulate air, *MDR* multidrug resistant, *PPE* personal protective equipment

Current WHO guidance (from June 2020) recommends for healthcare workers caring for suspected or confirmed SARS-CoV-2 positive the use of droplet and contact precautions, and airborne and contact precautions in case of aerosol generated procedures, although there are data supporting the universal use of airborne + contact precautions. Recent local and international guidelines should be followed for updated procedures on prevention of SARS-CoV-2 infection.

Upon entering HCT unit, *visitors* should be screened for the presence of symptoms of easily transmissible diseases such as viral respiratory tract infections, gastroenteritis, etc. and, if present, advised to postpone their visit until no longer symptomatic. Also, healthcare workers with respiratory symptoms should refrain from direct patient care until the symptoms resolve. Non-immune persons who were exposed to communicable diseases such as measles or chickenpox should refrain from contact with HCT recipients or transplant candidates until the incubation period passes without developing the disease.

Instructional materials for patients and visitors on recommended hand hygiene, respiratory hygiene/cough etiquette practices and the appli-

cation of transmission-based precautions should be provided. Vaccination of healthcare workers and household contacts is paramount and discussed in the dedicated chapter.

27.4 Management of the Threat of MDR Bacteria

In the era of increasing bacterial resistance, an important part of infection control deals with *prevention of colonisation and infection with MDR bacteria* (Siegel et al. 2007). Active surveillance, for example, with rectal swabs for detecting colonisation with carbapenem-resistant *Enterobacteriaceae* (CRE) or vancomycin-resistant enterococci (VRE) or nasal swabs for methicillin-resistant *Staphylococcus aureus* (MRSA) should be performed in institutions where these pathogens are encountered or in patients coming from such institutions.

The need for screening for different pathogens may vary according to local epidemiology and is performed in some settings (Girmenia et al. 2015), while recent ASTCT guidelines recommend that when CRE are uncommon, limiting screening to patients referred from CRE-endemic areas is another reasonable strategy (Satlin et al. 2021). The screening for MDR pathogens allows to choose the appropriate empirical therapy in case of neutropenic fever and to implement contact precautions to prevent nosocomial transmission (Girmenia et al. 2015). Careful evaluation of the possibility and risk/benefit ratio of decolonisation in selected cases through oral administration of non-absorbable molecules or faecal microbiota transplantation is warranted (Girmenia et al. 2015; Bilinski et al. 2017), although evidence is insufficient to provide a recommendation (Tacconelli et al. 2019).

In order to counteract the threat of MDR pathogens and the shortage of agents active against Gram-negative MDR bacteria, *antimicrobial stewardship program* should be implemented in every centre (Gyssens et al. 2013). Additionally, national systems for surveillance, with obligation of notification and recommendations for containment and infection control measures, should be put in place (Tacconelli et al. 2014).

The aim of *antimicrobial stewardship* is to limit the negative impact of MDR pathogens on patients' outcome, and its *main elements* are detailed in Table 27.2.

Successful implementation of antimicrobial stewardship is based on a multidisciplinary approach and close collaboration between the treating haematologists, microbiology laboratory and infectious diseases consultation service, and includes also infection control unit, hospital pharmacy and hospital authorities who should recognize that this is an important step in high-quality management of infectious complications after HCT.

Surveillance of effectiveness of infection control practices should be put in place, with regular

Table 27.2 Main elements of antimicrobial stewardship program

1. Regularly updated (e.g. every 6–12 months) surveillance of local epidemiology of infections in HCT recipients, through reports on: (a) Resistance rates to main antibiotics in top 10 most frequent pathogens (b) Data on antibiotic consumption (c) Data on patient outcomes in case of most frequent/difficult infections, particularly due to resistant pathogens
2. Implementation of updated diagnostic methods and prompt reporting of microbiologic results by the laboratory in order to provide clinicians with (a) Correct and timely diagnosis (e.g. of viral or fungal infections or *Clostridium difficile*, which may allow to avoid unnecessary antibiotic therapy) (b) Rapid results of antimicrobial susceptibility testing to allow choosing the best targeted antibiotic therapy
3. Promoting appropriate antibiotic use, for example (a) Implementing timely de-escalation or discontinuation of antibiotic treatment, particularly during neutropenia (b) Appropriate dosing for different indications (c) Optimised infusion strategies for time- and dose-dependent molecules, e.g. use of extended or continuous infusion of time-dependent molecules such as beta-lactams
4. Establishing and regularly updating protocols for prevention and treatment of infections, e.g. identifying antibiotic and antifungal regimens for empirical therapy in accordance with local epidemiology (e.g. prevalence of extended spectrum beta-lactamase (ESBL) producing *Enterobacteriaceae*, methicillin-resistant staphylococci, azole-resistant aspergilli, etc.)

monitoring of adherence. In case of contact-transmission pathogens, such as *Clostridium difficile* or MDR bacteria, laboratory data should be regularly analysed to detect any trends indicating possible increase in transmission.

27.5 HCT Environment

Flowers, fountains, water leaks and water-retaining bath toys carry the risk of *water-associated infections* with Gram-negative bacilli such as *Pseudomonas aeruginosa* or *Legionella* and thus should be avoided in the areas where severely immunocompromised patients are being cared for (Yokoe et al. 2009). In addition, there are issues specific for HCT recipients, such as room ventilation, intensified protective measures applied during hospital construction and renovations, avoidance of contact with soil (including potted plants) and avoidance of dust, both permanently (e.g. non-carpeting and no porous surfaces) and while cleaning, all aimed at decreasing the risk of invasive aspergillosis (Yokoe et al. 2009).

CIBMTR/ASBMT/EBMT global recommendations on protective environment concerning hospital *room design and ventilation* are available (Yokoe et al. 2009). Briefly, allo-HCT recipients should ideally be placed in protective environment rooms that incorporate several features including central or point-of-use HEPA (high-efficiency particulate air) filters with 99.97% efficiency for removing particles ≤0.3 μm in diameter and ≥12 air exchanges/hours, with directed airflow and consistent positive air pressure differential between the patient's room and the hallway ≥2.5 Pa. All these measures remove airborne fungal spores and are aimed at preventing airborne infections with filamentous fungi such as aspergilli. The efficacy of protective isolation measures in case of auto-HCT recipients is less well established.

Currently, *HEPA-filtered rooms* are available in almost all centres, while few centres fulfilled all the CIBMTR/ASBMT/EBMT requirements. However, the knowledge on the details and maintenance of protective environments in the HCT setting was recently found inadequate, requiring education efforts and cooperation with hospital infection control and the hospital maintenance services (Styczynski et al. 2018).

During *construction and renovations*, due to high density of fungal spores, protective environmental measures are particularly important, and mould-control measures should be intensified and filtration efficiency should be monitored frequently to best determine appropriate time for replacement. Specific recommendations are available and should be followed (Sehulster et al. 2004). For example, construction and renovation areas should have negative air pressure relative to HCT patient care areas to ensure that air flows from patient care areas towards construction areas, and a portable, industrial-grade HEPA filter should be used between a construction zone and the HCT unit if a large area is under construction and negative pressure differential cannot be guaranteed. In addition, HCT recipients may benefit from wearing N95 respirators outside HEPA-filtered areas, particularly during ongoing constructions, since unlike surgical masks, higher efficiency masks offer protection against *Aspergillus* spores. Active monitoring of cases of invasive mould infections should be performed in order to detect any possible outbreak.

27.6 Food Safety in Transplant Recipients

Drinking *water* should be safe; thus boiled or bottled water is preferred. Tap water in highly populated areas is usually regarded as safe from bacterial contamination because tested regularly. However, it may still contain *Cryptosporidium*. Water from private wells should be avoided.

The use of *low-microbial diet*, which prohibits fresh fruits and vegetables and unprocessed food, did not result in a decreased incidence of infections in neutropenic patients (Sonbol et al. 2015; van Dalen et al. 2016). Standard food safety practices that emphasize safe handling and washing or thoroughly cooking food were found to be just as safe and produced no increase in infection rates or incidence of neutropenic fever. Similarly,

to other immunocompromised patients, HCT recipients should avoid foods possibly contaminated by *Listeria monocytogenes*, *Campylobacter jejuni*, *Salmonella enteritidis*, *Toxoplasma gondii*, etc.

Main high-risk foods to avoid include:

- Raw or undercooked meat, poultry, fish or shellfish
- Refrigerated smoked fish
- Unpasteurised milk
- Foods with raw or undercooked eggs
- Unwashed fruits and vegetables
- Raw sprouts
- Soft cheeses made from unpasteurised milk like brie, camembert and blue-veined and fresh cheese (can be eaten if cooked)
- Hot dogs, deli meats and luncheon meats that have not been reheated to steaming hot or to 75 °C
- Unsafe water and ice made of it

Food safety practices for food handling should be followed, and specific information for cancer patients is available online (https://www.fda.gov/downloads/Food/FoodborneIllnessContaminants/UCM312793.pdf). It should be kept in mind that too restrictive diet recommendations, in the absence of the clear benefit of avoiding foods other than above-mentioned, may have negative impact on patient's nutritional status and/or quality of life.

Key Points

- General guidelines for preventing healthcare-associated infections should be followed, and hand hygiene is the single most effective measure.
- Mandatory isolation procedures comprise Standard Precautions and Transmission-Based Precautions if appropriate: airborne, contact or droplets.
- Specific recommendations on ventilation, room design and protective environment during construction/renovation are provided to protect HCT from transmission of spores of filamentous fungi, mainly *Aspergillus*.
- Protocols for the prevention of colonisation and infection with multidrug-resistant bacteria should be put in place, particularly in centres where these bacteria are already present.
- Antimicrobial stewardship program should be implemented in every centre to promote optimal use of antibiotics.
- Standard food safety practices should be applied, and only selected foods should be avoided (e.g. raw/undercook/underheated meat, fish or eggs, unpasteurised milk, unwashed fruits and vegetables, unsafe water).

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